

Contextual variation supplants cultural stereotypes and destabilizes reasoning in LLMs

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With large language models (LLMs) being increasingly deployed in high-stakes settings, concerns about their safety and reliability across contexts are becoming ever more pressing. A central thread in the model safety literature examines harmful gender and social biases, with many studies showing correlations between LLM behavior and cultural stereotypes. Most evaluation procedures, however, rely on decontextualized single-sentence probes. This assumes that biases measured in one setting generalize across contexts—an assumption at odds with linguistic theory, which understands language as inherently contextually-dependent. We thus develop an experiment to test whether gender biases in LLMs remain stable amidst minimal contextual variations. Our results show that when context is introduced in the form of a priming sentence which is discourse-relevant but uninformative towards the task, biases shift and most models cease to show significant correlations with cultural stereotypes; instead, model behavior is best predicted by seemingly innocuous features in the prompt and priming sentence. An analysis using Contextuality-by-Default (1) further reveals inference patterns which diverge from how humans are known to reason based on linguistic context, and which signal a deep, structural unpredictability in biases. These results expose the limitations of decontextualized bias evaluations, raise concerns for safety assessments and real-world deployment, and offer new insights into the pragmatic and reasoning capacities of LLMs.

Keyword 1 | Keyword 2 | Keyword 3 | ...

As AI tools continue to proliferate into numerous facets of everyday life, concerns regarding their behavior in implementations continue to mount. Existing work in the model bias literature has shown that even state-of-the-art (SOTA) large language models (LLMs) can exhibit harmful biases towards already-marginalized communities (2–8). Addressing this problem has been noted as a key challenge by UNESCO, who has recommended that “AI actors should make all reasonable efforts to minimize and avoid reinforcing or perpetuating discriminatory or biased applications and outcomes throughout the life cycle of the AI system to ensure fairness of such systems” (9).

Because model biases often correlate with societal biases (10–14), it is thought that these biases “leak” into the model during training, suggesting that models are reasoning on the basis of some broader cultural context (10). In most experimental settings, however, model bias is evaluated in isolation, typically using decontextualized single sentences or utterances (15–18). While this removes confounding variables in an already highly nonlinear system, it may compromise the generalizability of these findings to realistic, context-rich interactions.

Natural language use inherently relies on abductive reasoning based on contextual cues (19). It has been shown, however, that LLMs struggle to perform this type of reasoning (20). Consequently, whether language-model biases are robust to contextual variation remains unresolved. It is also unclear whether models continue to rely on cultural context in their reasoning—as they appear to in decontextualized settings—or whether their reasoning strategies shift once additional discourse context is introduced.

Significance

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To address these questions, we devise a simple experiment to investigate how the addition of discourse context affects the way that SOTA LLMs reason about an ambiguous grammatical feature. Specifically, we ask models to infer the undetermined gender of a referent in a forced-choice task. We test six models by running this experiment on 360 sentence templates in five context settings. Results are analyzed by estimating the probability that each model will generate the feminine pronoun in each context setting for each sentence template.

Comparing our generation probability estimates with empirical data on gender stereotypes confirms the presence of biases in the production of gendered pronouns that are significantly correlated with cultural stereotypes. However, we find that these biases are inconsistent across contextually-equivalent settings, and correlations with stereotypes disappear when seemingly innocuous context is introduced. An analysis of the mutual information between various linguistic features and model output further reveals that when discourse context is present, biases in most models are better predicted by irrelevant gender features in the context than by cultural stereotypes. Finally, we assess whether models' responses merely correlate with the presence of gender features in the context, or whether they exhibit genuine contextual dependence. To adjudicate between these possibilities, we apply the Contextuality-by-Default framework (21). Our analysis reveals several instances of contextuality, suggesting that the observed biases often cannot be reduced to simple feature correlations across conditions.

Altogether, these results show that context introduces a fundamental source of unpredictability into model behavior: biases observed in one context do not reliably generalize to even minimally altered contexts. As such, we conclude by considering the implications of these findings for the evaluation of gender bias in LLMs, research on the pragmatic capacities and inference strategies of modern LLMs, and the application of AI in high-stakes domains such as medicine, law, and politics.

Estimating Pronoun Generation Probability. We begin by estimating the generation probabilities for each sentence template in the WinoPron Schemas (22) across five "context settings". We consider the first context setting to be the unprimed setting, c_0 . In this setting, there is only one sentence template with a BLANK in the place of a pronoun with an unambiguous antecedent. We refer to the template as the "target sentence" and to the pronoun being inferred by the model—our key observable—as the "target pronoun." We then consider two settings, c_f and c_m , in which there is a priming sentence added before the target sentence that adds discourse context. This priming sentence has the same discourse referents as the target sentence and includes either a feminine or masculine pronoun, respectively, whose antecedent is the *other* discourse referent and *never* the same referent as the target pronoun. We refer to the pronoun in the priming sentence as the "priming pronoun". It is important to note that, in all cases, c_m and c_f differ only by a single word. Finally, we consider two null settings, $c_{0,1}$ and $c_{0,2}$, in which there is a priming sentence with no pronouns and which is irrelevant to the target sentence.

For each sentence template, we estimate the probability of generating the feminine pronoun in each context

setting. Because we filter responses to only accept trials in which the model responds with a gendered pronoun (see Materials and Methods), we are able to treat this as a Bernoulli process. Thus, if we let the set of templates in the dataset be $T = \{0, \dots, 359\}$ and adopt the convention that $\hat{p}_i(1|c) \sim P(X_i \in \{\text{she, her, hers}\} | C_i = c)$ is the maximum likelihood estimate for some template $i \in T$ and context setting, c , then $\hat{p}_i(1|c)$ is simply the fraction of runs in which the model generates the feminine pronoun.

Figure 1a-f shows the distribution of empirical generation probabilities for all templates in the dataset. The distributions shown for c_0 represent the baseline bias of these models. Looking at this curve for GPT OSS 20B indicates that the values of $\hat{p}_i(1|c_0)$ for all $i \in T$ are approximately normally distributed around 0.5. So, for most templates, in the dataset, there is approximately a 50% chance that the model will select the feminine pronoun. This would suggest that GPT OSS 20B exhibits very little bias in the unprimed setting, while Phi-4, Gemma 3 27B, and Qwen 2.5 7B all exhibit a bias towards the masculine pronoun. Conversely, our findings for the Llama models confirm those of (5) in that there is a slight preference for the feminine pronoun in the unprimed setting.

Looking at the distributions of $\hat{p}_i(1|c_f)$ and $\hat{p}_i(1|c_m)$ for all $i \in T$, it is immediately clear that the gender marking on the priming pronoun can radically alter generation probabilities for the target pronoun in all models tested. Furthermore, changes due to the introduction of discourse context are clearly not uniform across models (or, in the case of Llama, even model families). While the presence of a male pronoun in the priming pronoun appears to change very little compared to the unprimed setting in most models, GPT OSS 20B appears to suddenly strongly prefer the masculine pronoun as the target pronoun, and Llama 3.2 1B moves in the opposite direction, slightly increasing its preference for the feminine pronoun. In general, generation probabilities appear to shift in the direction of the priming pronoun for all models except Llama 3.2 1B, which exhibits the opposite behavior. Looking to the null settings, some salient shifts appear compared to the unprimed case—especially for the two Llama models—but in general, these distributions appear to shift considerably less than c_f and c_m .

Looking now to Figure 1g-l, we find that in all except the Llama models, c_f and c_m diverge considerably more from the unprimed setting than the two null context settings, on average. We assess this by calculating the Kullback-Liebler Divergence, $KL_i(c \| c_0)$ as a measure of how much the distribution of responses in setting c differs from the distribution of responses in the unprimed setting for template i (see Materials and Methods). For ease of comparison, we report this as the average divergence, $\langle KL(c \| c_0) \rangle = \frac{1}{|T_v|} \sum_{i \in T_v} KL_i(c \| c_0)$, where $T_v \subseteq \{0, \dots, 359\}$ denotes those templates for which valid measurements were collected in all contexts for that model (see Valid Measurement Count Supplement). We find, moreover, that in all models tested, generation probabilities diverge most from the unprimed context setting when the discourse context introduces a feminine pronoun. Considering the absolute value of the divergence, three classes of models appear to emerge. The first is those models in which c_m and c_m diverge considerably from c_0 , while the null see little change. In this class, we

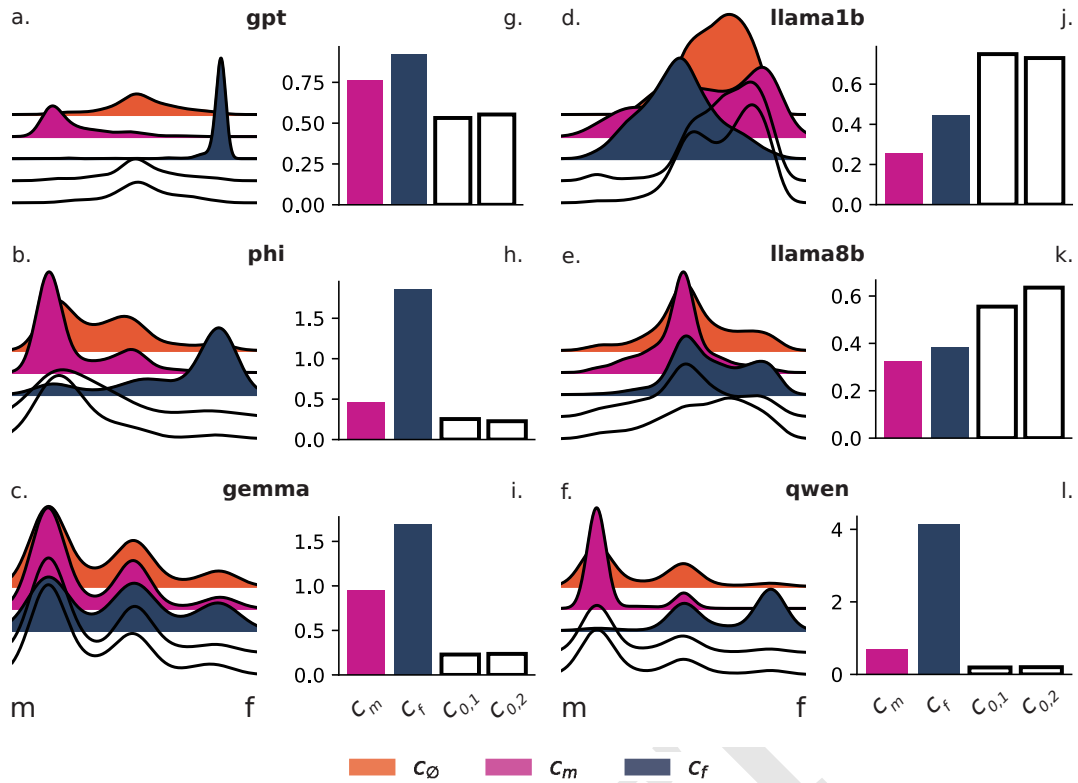


Fig. 1. (a-f) Distribution, over all templates, of empirical generation probabilities. The height of the curve at the left extrema of each joy plot indicates the fraction of templates for which, in that context setting, $\hat{p}(0|c) = 1$ (the probability of generating the masculine pronoun in context setting c). The inverse is true for the height of the curve at the right extrema, so that this indicates the fraction of templates for which $\hat{p}(1|c) = 1$ is the right extrema. (g-l) Average Kullback-Liebler Divergence $\langle \text{KL}(c \| c_0) \rangle$ (in bits) between the distribution of responses for each template in the unprimed setting, c_0 , and one of the primed settings tested, $c \in \{c_f, c_m, c_{0,1}, c_{0,2}\}$.

include Phi-4, Gemma 3 27B, and Qwen 2.5 7B, whose average KL divergence for the two discourse context settings ranges from 0.47 bits - 4.15 bits, while the two null context settings are tightly bound from 0.19 bits - 0.26 bits. Opposite to this are the two Llama models, in which divergence in the discourse context settings is slightly less than that of null settings. In these models, divergence between c_0 and either c_f or c_m is limited to the range 0.36 bits - 0.44 bits, while divergence between c_0 and $c_{0,1}$ or $c_{0,2}$ is slightly higher at both extremes, ranging from 0.55 bits - 0.75 bits. Finally, GPT OSS 20B, like the first group, exhibits more divergence between the unprimed and discourse context settings, ranging from $\langle \text{KL}(c_m \| c_0) \rangle = 0.76$ bits in the masculine setting to $\langle \text{KL}(c_f \| c_0) \rangle = 0.92$ bits in the feminine setting, but difference between this divergence and that between unprimed and null context is not as stark, with $\langle \text{KL}(c_{0,1} \| c_0) \rangle = 0.53$ bits and $\langle \text{KL}(c_{0,2} \| c_0) \rangle = 0.55$ bits. In all models except Llama 3.1 8B, the difference $\langle \text{KL}(c_f \| c_0) \rangle - \langle \text{KL}(c_m \| c_0) \rangle$ was greater than the difference $|\langle \text{KL}(c_{0,1} \| c_0) \rangle - \langle \text{KL}(c_{0,2} \| c_0) \rangle|$.

Correlations with Cultural Stereotypes. Of the 360 templates in the WinoPron dataset, the antecedents of the target pronouns of 279 templates were able to be matched with a role noun (i.e. plumber, student, child) in (23). Thus, using the empirical cultural gender stereotype ratings measured in (23), we were able to find significant Spearman correlations between femininity judgements about the antecedent of the target pronoun (which is always a role noun) and the empirical generation probability of the female pronoun in the unprimed setting

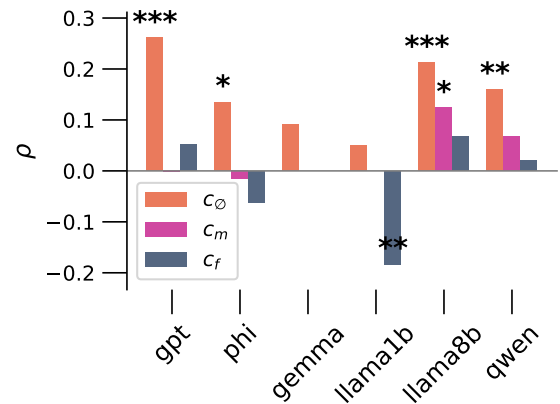


Fig. 2. Spearman Correlation (ρ) between empirical generation probability from our experiments and cultural gender stereotypes as measured by (23), with significance (***) := $p < .001$, (**) := $p < .01$, (*) := $p < .05$

for all models except Gemma 3 27B and Llama 3.2 1B. However, as shown in Figure 2, these correlations both decrease and vanish in most cases when priming sentences are introduced. In this case, most models do not show any significant correlations between femininity judgements and empirical generation probability of the feminine pronoun. The only exceptions to this come from Llama family models, which show a correlation between cultural stereotypes and pronoun selection in the fem-primed setting in Llama 3.2 1B, and a correlation between cultural stereotypes and pronoun selection in the masc-primed setting in Llama 3.1 8B. In the latter case, the correlation is both weaker and less significant.

This finding is also replicated when we move from the singular, directional correlation analysis between cultural stereotypes and generation probabilities to a comparative, nondirectional analysis across contextual features. We consider five linguistic features of the text: (1) if present, the gender marking on the priming pronoun, (2) the type of role (either occupation or participant) of the antecedent of the priming pronoun, (3) cultural stereotypes (i.e. empirical femininity ratings from (23), (4) the grammatical case of the pronoun, and (5) the linear order of options presented in the prompt (see Materials and Methods). For analytical ease, we group (2) and (3) together as "cultural contextual features", and we group (4) and (5) together as "mere syntactic features" because they bear no contribution to the semantics.

Figure 3 shows the distribution, over templates, between model output (i.e. the pronoun selected) and each feature tested. We find that while cultural contextual features contribute some information towards predicting the selected pronoun, this contribution is generally less than the contribution from mere syntactic features in all models tested (see Figure 3, left). Once a priming sentence is introduced, however, we find that the gender of the priming pronoun, in most templates, provides more information towards model output than any other feature when using Phi-4, Qwen 2.5 7B, and especially GPT OSS 20B. In Gemma 3 27B, margins are tighter, but the gender of the priming pronoun is more informative than all other features, on average. Looking at Llama 3.1 8B or Llama 3.2 1B, the priming pronoun is still, on average, more informative towards the model's selection for the target pronoun. However, in these, the average amount of information gained by knowing linear order is more than that of the average amount of information gained by knowing the priming pronoun.

Contextuality-by-Default. While the analysis thus far has allowed us to make claims about how changing the priming pronoun affects model behavior statistically, distributional assessments do not allow us to determine whether these shifts reflect a deeper dependency on the priming pronoun itself or merely the influence of latent variables shared across conditions. Because even null primes shift distributions, statistical effects alone cannot establish that the priming pronoun itself determines the response, rather than merely correlating with some broader contextual factor. By applying the Contextuality-by-Default framework, we test whether probabilities for the target pronoun can be regarded as fixed prior to exposure to the prime and *shifted* based on its gender, or whether probabilities for the target pronoun becomes well-

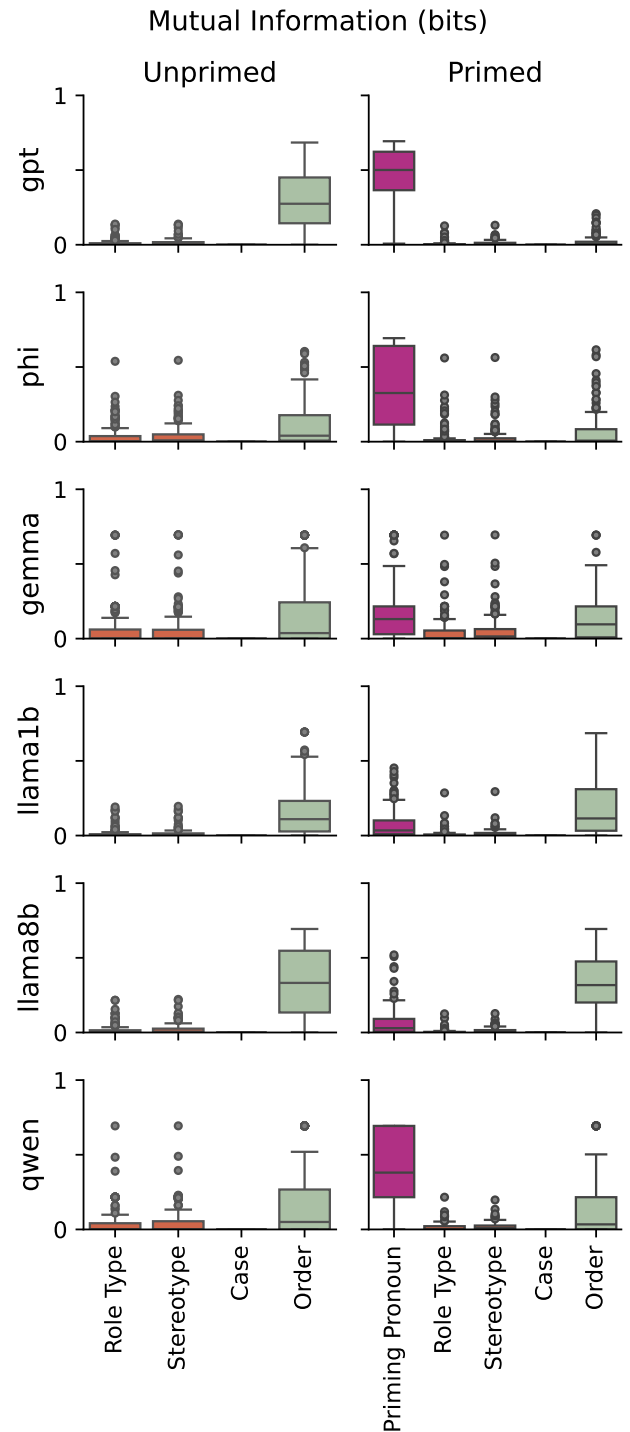


Fig. 3. The mutual information, measured in bits, between features of the prompt in unprimed context settings (left) and primed context settings (right) and the pronoun generated by the model. Role Type and Stereotype refer to cultural gender biases with regard to subjectivity, occupations, and discursive roles. Case and Order refer mere syntactic features including pronoun case and the linear order of pronoun options presented in the prompt. Priming Pronoun (only in primed context settings) refers to the gender of the irrelevant pronoun in the priming sentence.

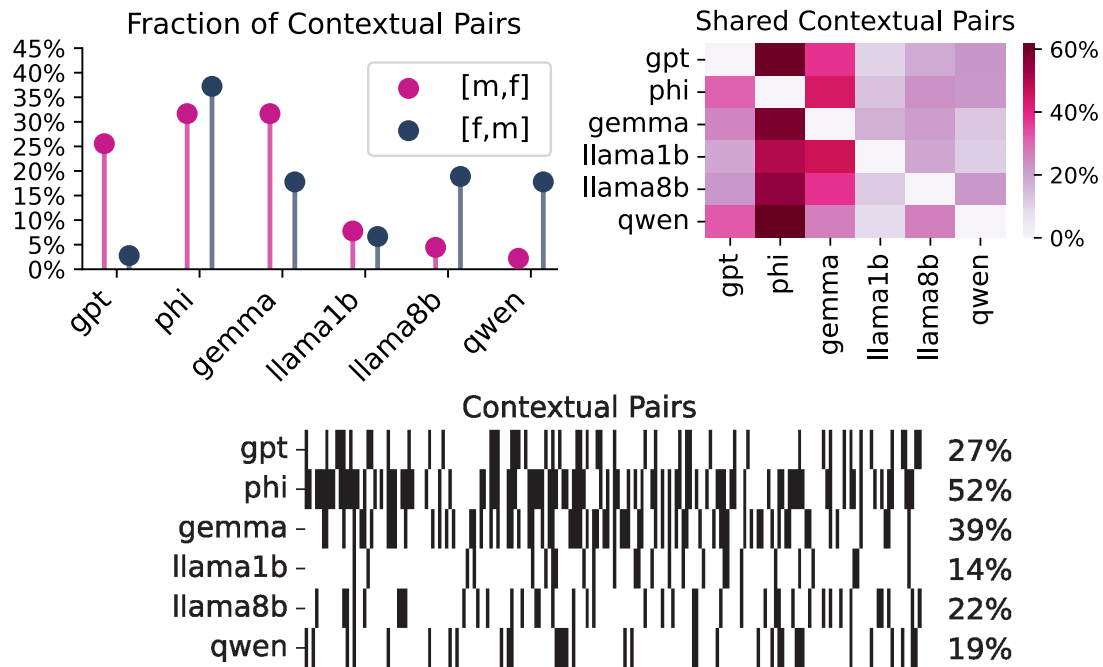


Fig. 4. (top left) Fraction of template pairs which exhibited contextual measurements, disambiguated by the linear order of pronoun options presented in the prompt. (top right) Number of template pairs which exhibited contextual measurements in both models. (bottom) For each model across all template pairs, a black bar indicates that in that model, that template pair exhibited contextual measurements in either the linear order of pronoun options. The value on the right of each row indicates what fraction of templates exhibited contextual measurements in either order for that model.

defined only in relation to the specific priming pronoun encountered. Because assessing contextuality requires measuring both generation probabilities for both the priming and target pronoun, it is defined to pair of templates, not individual templates. If contextuality is detected for a pair of templates, it implies that the model is explicitly dependent on the priming pronoun to infer the target pronoun in one or both templates (see Materials and Methods).

Figure 4(bottom) shows that in all models, several pairs exhibit contextuality. A cursory visual inspection suggests, however, that there is little overlap between models as per which template pairs exhibited contextual measurements. This is confirmed by Figure 1(top right), which shows that overlap is high with Phi-4, in which the majority of templates exhibit contextual measurements, but falls below 40% for most other model pairs. Finally, we find that contextuality measurements differ based on the linear ordering of options presented. This is most stark in GPT OSS 20B, in which more than a quarter of pairs exhibited contextual measurements in the (masc pronoun, fem pronoun) ordering, while in the (fem pronoun, masc pronoun) ordering exhibited contextual measurements in only 3% of template pairs. Alternatively, the opposite is true for Qwen 2.5 7B, in which contextuality was detected in 2% of pairs when in the former ordering and 18% of pairs in the latter.

Discussion

Gender bias has been especially well-studied in the language modeling literature. It has been the topic of a several papers (16, 24, 25) and workshops (26, 27). Countless benchmarks (28–33) have been created to assess the presence of bias. Most of these benchmarks and studies on bias, however, rely on tasks in which utterances are

stripped of their discursive context (15–18). The present study attempts to go beyond this limitation to understand how context affects gender biases in language models.

Our results provide several insights into how contemporary LLMs process gendered anaphora under varying contextual conditions. In interpreting these results, it is helpful to distinguish between context and “co-text”. The latter has historically been used to distinguish between the broader setting in which a sentence is uttered (context) from the surrounding text which may influence a sentence’s interpretation (co-text) (34, 35). With this distinction in mind, we can frame this study as an attempt to understand the contribution of *co-textual* features on the gender bias exhibited by modern LLMs, as compared to the well-documented contribution from *contextual* features (specifically, cultural gender stereotypes).

A visual comparison of Figure 1a–f makes it immediately evident that models’ gender biases shift drastically when co-text in the form of a priming sentence is introduced. This is true even for Gemma 3 27B, a model which has been trained on the dataset from which our templates originate CITE . Furthermore, the shift observed is dependent on aspects of the introduced co-text, and differs considerably across models. Despite the co-text providing no information towards the inference in any instance, it is clear that when the priming sentence includes the same discourse referents as the target sentence, then for most models, the shift in bias appears to be more pronounced than it is in the null setting, where the priming sentence bears no relation at all to the target sentence. This shift depends, moreover, on the gender of the pronoun in the priming sentence, even though this pronoun is theoretically uninformative towards the gender of the pronoun being inferred.

This is confirmed by an analysis of divergence between primed context settings and the unprimed context setting. When the priming pronoun is masculine, all models tested behave more similarly to the unprimed case than when the priming pronoun is feminine. Moreover, the difference between these settings is often stark, especially as witnessed in Qwen 2.5 7B. Comparison of this difference with the difference between the two null context settings suggests that, for all models except Llama 3.1 8B, the minimal change of swapping the gender of the priming pronoun induces a systematic change in bias. Altogether, these shifts in bias observed between unprimed and primed settings—and especially between contextually equivalent sentences—suggests that findings regarding not only the magnitude, but also the direction of model biases may be fragile and heavily dependent on seemingly innocuous linguistic features, as opposed to a learned and semantically-grounded bias.

Having established that co-text can modulate model behavior, we move on to comparing this to the effect had by cultural context. Spearman correlations observed in the unprimed setting confirm previous findings that when no additional context is present, cultural gender stereotypes are significantly correlated with models' gender biases (12–14). In this decontextualized setting, we also validate results from (author?) (10) that cultural gender stereotypes influence gendered referent selection when the referent is underdetermined. The disappearance of these correlations once additional discourse-relevant co-text is introduced, however, suggests that models do not treat cultural stereotypes as fixed priors, but as cues that are easily overridden. Furthermore, it has been asserted in the literature that, when tasked with reasoning about underdetermined referents, models rely on cultural stereotypes (10, 31). While our results in the unprimed setting attest to this, the picture is muddled by fact that the primed settings theoretically provide no additional information, but correlations with cultural stereotypes vanish in most models. This would suggest that (b) models' reasoning about the gender of referent may not only depend on the information they have about the referent, but also theoretically irrelevant contextual information which can take precedence over any cultural stereotypes they may fall back on.

Considering how informative cultural stereotypes are towards model bias compared to other features, we find that across all models, even no co-text is present, this cultural context reduces uncertainty in pronoun selection by less than 0.1 bit on average across all models. In all models, cultural stereotypes were less informative of pronoun selection on average than linear ordering of options in the prompt. In most models, this linear order selection effect was also considerably less informative than the gender of the priming pronoun, when discourse-relevant co-text was introduced. Altogether these results appear to confirm the hypotheses in that models' gender biases may not be as grounded in cultural stereotypes, and more dependent on innocuous co-textual features than initially imagined.

Our analysis of contextuality provides further insights into our hypotheses by determining whether the changes in generation probabilities observed are the result of some fixed, prior bias that may be due to cultural stereotypes and *shifted* by the gender of the priming pronoun, or if

this bias is not fixed at all, but rather constructed contextually explicitly on the basis of this priming pronoun. The presence of contextuality across several instances in all models suggests that in these cases, it is the latter.

Some work on contextuality in human decision making has claimed that sentence-order changes in surveys can invoke contextuality in human responses (36, 37). This, however, has been shown not to be the case. The apparent contextuality was reported due to a misapplication of the sheaf-theoretic calculations of contextuality, which assume marginal selectivity (fixed margins over all conditionals) (21, 38). Prior work in a linguistic priming experiment has furthermore confirmed the lack of contextuality in human judgements in this setting (39).

Taken together, these findings signal not only a deep instability in model biases, but raise important questions about the divergence between how humans and language models reason about gender, as well as possibly other pragmatic features of language.

All of the results of this study suggest that models may be differentially prone to conflating discourse referents. For example, consider this primed template from the dataset, with bindings made explicit

- (1) *The mechanic₁ called to inform the customer₂ that pro₂ car would be ready in the morning. The mechanic called to inform the customer that BLANK₁ had completed the repair.*

Here, we use “pro” as a stand-in for the pronouns his or her. While it has been shown repeatedly that humans readily assume the gender marking on the target pronoun, BLANK₁, by deploying gendered stereotypes about mechanics CITE Carreiras, there appears to be no evidence that one would “reason across indices” in such a way that their inference regarding the gender marking on BLANK₁ relies on the gender marking on pro₂. Nevertheless, this is what all models tested appear to do in several instances. Theoretically, as abductive reasoning is the cornerstone of pragmatics (19, 40), this pragmatic failure appears to lend further evidence to the more general finding that SOTA language models exhibit poor performance in abductive reasoning tasks (20),

Our results show that context introduces a fundamental source of unpredictability into model behavior: biases observed in one context do not reliably generalize to even minimally altered contexts. This has immediate implications for the evaluation of model bias, suggesting that the model bias literature must take contextual effects into account if its conclusions are to remain robust. Biases measured in decontextualized settings may not generalize to richer linguistic environments, undermining the reliability of many standard evaluation protocols. This has serious implications for safety testing, which frequently relies on decontextualized or minimally contextualized prompts to diagnose harmful behavior. The inconsistency we observe across context settings raises concerns for systems already deployed in high-stakes domains, where unpredictable contextual shifts may lead to inconsistent or inequitable outcomes.

Our results also contribute to the growing literature on the pragmatic capacities of LLMs. Where many studies continue to tout improved linguistic capabilities and human-like behaviors in newer models, our findings

suggest a deviation between humans and AI in one of the most fundamental aspects of language use—situated reasoning. This, in turn, also invites further theoretical reflection on distributional semantics as a model of meaning: if the distributional patterns underlying LLMs induce pragmatic behaviors that humans do not exhibit, then strengthening the bridge between distributional semantics and the formal study of anaphora may yield key insights.

Finally, our results contribute to the literature on gender bias by demonstrating that model biases can be reshaped—or, in many cases, overridden—by discursively relevant but theoretically inconsequential context. This raises concerns for high-stakes domains such as medicine, law, and politics, where unpredictable contextual sensitivity can produce inconsistent or harmful outcomes. Together, these findings highlight the need for bias-evaluation methodologies that account for context-dependent behavior and for governance frameworks that recognize the risks posed by deeply unstable reasoning in deployed LLMs.

It should be noted that the present study is only a first step in understanding context-dependent reasoning in language models and thus, limited in several ways. First, due to cost constraints, we do not test the largest proprietary API models, so it remains possible that these models would display different patterns of bias and reasoning. Also due to cost, computational constraints, and environmental consideration, we limit the number of runs of each model in each context setting despite the fact that additional runs would offer a finer-grained characterization of the underlying distributions. Especially because we only select instances in which models responded appropriately according to the template, additional runs would likely significantly improve our estimations of generation probability. Finally, although our experimental setup introduces realism through the addition of relevant discursive context, it still does not fully capture the complexity of real-world usage, where discourse structure, style, and user interaction may introduce further variability. Future work should address these considerations and begin investigating sources of variance in these models, especially in those in the Llama family. Furthermore, establishing a human baseline would be invaluable for better comparison between human and LLM gender biases and reasoning. Finally, while this study focuses primarily on generation, all of the tools have been developed to accommodate the same study at the level of next token prediction, without a generation head. A comparative analysis of how these biases align with biases in generation probabilities may unveil key insights into the sources of bias and instability.

Materials and Methods

Models. We test some of the most popular (by recent downloads) LLMs currently available on the open-source platform HuggingFace. All LLMs except Gemma 3 27B were ran locally at 4-bit quantization using H200 GPUs (see [Valid Measurements/Experiment Supplement](#) for further details). Gemma 3 27B was run unquantized due to known issues with quantization and the Gemma 3 tokenizer*.

All analyses presented were run with the following fixed generation parameters: a temperature of $T = 0.5$, `max_new_tokens = 6` and `top_k =`

40. For robustness, all analyses were recreated for a selection of models at temperatures $T = 0.3$ and $T = 0.7$. These results can be found in Supplement.

Dataset. To assess model biases in pronoun completion, we adopt the WinoPron schemas (22), a recent modification of the earlier WinoGender schemas (41). The WinoPron schemas are a set of 180 paired sentence templates (360 templates total) written in English. Each template has two agents, each with a role noun. In every template, one role noun is that of an occupation (i.e. technician, nurse, librarian) while the other is a “participant” (i.e. taxpayer, customer, child). Templates in the dataset come in occupant-participant pairs such that both templates will have the same occupant-participant roles. The two templates in an occupant-participant pair differ by which role acts as the antecedent of the pronoun, of which there is always one. The WinoPron schemas have been designed furthermore to enable argument agreement for singular they and neopronouns, and the grammatical case of pronouns is balanced across the dataset.

Both the WinoGender and WinoPron schemas were originally developed as pronoun resolution tasks to test models’ ability to reason about anaphora. We, however, have adapted them to function as a pronoun *selection* task. This is done simply by fixing the occupation and participant role variables and replacing the pronoun variable with the token BLANK as a masked token for the model to resolve (see [Prompting Supplement](#)).

Experimental Procedure. We employ a forced-choice experiment which proceeds by prompting models to select one of two gendered pronouns to replace the BLANK in the target template (see [Prompting Supplement](#)). This allows us to assess the biases of models because in every case, there is a singular, unambiguous antecedent of undetermined gender. A perfectly unbiased model would thus be expected to produce the masculine and feminine pronoun with equal probability.

By “unambiguous antecedent” we mean that, for example, the sentence (2) would not be permissible because the pronoun could refer to either agent.

(2) *The customer asked the worker if BLANK could leave the room.*

A more subtle crucial consideration in this experimental setup is that it requires possible completions to be contextually equivalent (42) up to gender presupposition. By this, we mean the following.

Consider, for example, the sentences:

- (3) a. *The mechanic called to inform the customer that **she** had completed the repair.*
- b. *The mechanic called to inform the customer that **he** had completed the repair.*
- c. *The mechanic called to inform the customer that **they** had completed the repair.*

If the mechanic in question identifies as a woman, then only is (3)a considered to a “felicitous” sentence. This is because the gender of a pronominal referent is considered a ϕ -feature (43, 44). A ϕ -feature is a part of the semantics of a sentence which constrains its interpretation on the basis of context or presupposition. Thus, without any evidence for the gender of the mechanic stated (3)a and (3)b are both structurally equivalent as well as contextually equivalent modulo ϕ , where we operationalize contextual equivalence as is done in (43, 45). Meanwhile, (3)c makes no gender presuppositions. This makes it both less informative, and defined in every context—therefor, it is not contextually equivalent to (3)a and (3)b. For this reason, we restrict models to selecting between male and female pronouns only. The goal here is not to exclude any gender identities, but quite the opposite. The goal is precisely to assess exactly how the prescriptive norms of the dominant gender binary might be encoded in language models as biases, despite researchers’ best efforts. Usage of non-binary pronouns prevents such an analysis by making no commitments to the gender of the referent. As such, a non-binary pronoun would theoretically always be the preferred selection for the model, telling us nothing about its reasoning strategies, or the emergence of gender biases.

To account for linear ordering biases (46, 47), we run half of the experiments by prompting the model to select from one of (`masc pronoun`, `fem pronoun`) while the other half is run with the ordering (`fem pronoun`, `masc pronoun`).

We introduce context into the experiment as a priming sentence. In settings c_m and c_f , the priming sentence is a target template’s occupation-participant pair. In c_f , the BLANK in the priming sentence is fixed as the feminine pronoun, while the opposite is true for c_m . In the null settings, the priming sentence is fixed as “The sky is blue.” for $c_{0,1}$ and “North is south.” for $c_{0,2}$.

In all experiments, only trials in which either the masculine or feminine pronoun was generated with proper formatting were recorded as data points. All other responses were discarded (see [Valid Measurement Count Supplement](#)).

Kullback-Liebler Divergence. Treating generation of the feminine pronoun for each i, c pair as a Bernoulli process, the KL Divergence between the distribution of responses observed in setting c and the distribution observed in c_0 takes the form

$$\text{KL}_i(c \| c_0) = p_i(1|c) \log \frac{p_i(1|c)}{p_i(1|c_0)} + (1 - p_i(1|c)) \log \frac{1 - p_i(1|c)}{1 - p_i(1|c_0)}$$

for all templates $i \in T_v$.

Calculating Stereotypical Correlations. To assess the extent to which cultural stereotypes may be informing model biases, we leverage a dataset of gender stereotypes in the form of femininity ratings for roles and occupations in English (23). We choose this as opposed to real data on gendered labor statistics because exactly half of the roles tested are not occupational, and because it was shown that social biases surrounding gender exhibit stronger correlations with model biases than labor statistics (10).

Mutual Information. The mutual information between contextual features and model responses for each template, i , was estimated as $I[X_i; F_i^c]$. Here, X_i denotes the the pronoun selected by the model to complete the target sentence, and F_i is the value of the feature. For example, where $F := \text{gender of the priming pronoun}$, $F_i^{cm} = 0$ for all i , and $F_i^{cf} = 1$ for all i . Alternatively, where $F := \text{cultural gender stereotypes}$, $F_i^c \in [0, 1]$ is the femininity rating for the antecedent of the target pronoun in i , for all context settings, c .

We elect for use of mutual information as a measure because it allows us to compare contextual features that are defined on different spaces and may exhibit differing, non-monotonic, and non-linear relationships with empirical generation probabilities. Unlike a correlation, which relies on our estimates of empirical generation probability, our mutual information analysis employs the `mutual_info_classif()` method from `scikit-learn` (48) to estimate entropy from k-nearest neighbor distances using all the individual measurements taken (49).

Contextuality-by-Default. Contextuality is an algebraic method which been used to study the behavior of LLMs in the face of anaphoric pronouns (50, 51), as well as garden-path sentences (52).

In this study, we elect to employ the Contextuality-by-Default framework over sheaf-theoretic contextuality specifically to avoid the issue of marginal selectivity in (36) that was discovered by (21). Because our measurements violate marginal selectivity, we operationalize the CbD approach outlined in (21), in which the authors create a CbD adaptation of the so-called "quantum question" (QQ) equality in (36). The QQ equality considers two questions, q_1, q_2 , with measurable random variables, X_1, X_2 , respectively. There are also two possible orderings, o_1 , and o_2 , where we let $o_1 := (q_1, q_2)$ and $o_2 = (q_2, q_1)$. With this setup, the degree of contextuality can be stated as

$$\Delta C = |\langle X_1^{o_1} X_2^{o_1} \rangle - \langle X_2^{o_2} X_1^{o_2} \rangle| - (|\langle X_1^{o_1} \rangle - \langle X_1^{o_2} \rangle| + |\langle X_2^{o_1} \rangle - \langle X_2^{o_2} \rangle|)$$

where

$$\langle X_i^{o_n} \rangle = 2P(X_i = 1|o_n) - 1$$

and

$$\begin{aligned} \langle X_i^{o_n} X_j^{o_m} \rangle = & 4P(X_i = 1|o_n)P(X_j = 1|o_m) \\ & - 2P(X_i = 1|o_n) \\ & - 2P(X_j = 1|o_m) + 1. \end{aligned}$$

The CbD equivalent of the QQ equality can thus be expressed as

$$\Delta C > 0, \quad [1]$$

which defines the conditions for contextuality. When Eq. 1 is satisfied, it means that there is an irreducible dependence between X_1 and X_2 . For the purposes of this study, this means that for a *pair* of sentences, the pronoun, X_1 , generated for sentence q_1 (referring to character *A*) depends

on the pronoun, X_2 in sentence q_2 (referring to character *B*), in such a way that any attempt to reduce the measurement of X_i as being independent of X_j renders an invalid joint probability space, based on the empirical measurements.

We further adapt this procedure to be facilitate analysis not only in generation settings, but also in next token prediction. This is done by employing a procedure which, in the quantum computing literature, is sometimes referred to as "steering" or as an "entanglement-verification test" CITE . This is simply the case where, instead of measuring both pronouns simultaneously, we measure the target pronoun (X_1 or X_2 , whichever is the pronoun corresponding to the second sentence in either of o_1 or o_2), while fixing the priming pronoun. This allows us much greater control when using causal models, as it allows us to capture both NTP probabilities and empirical generation probability estimates, and it allows us to do so using only one prompt—eliminating the complications and additional degrees of freedom introduced by multi-turn interactions.

[metaprompting supplement](#)

Data, Materials, and Software Availability. All code and scripts used in this study are available [here](#). The WinoPron Schema (22) is publicly available at their [project repository](#). Data on cultural gender stereotypes can be found in the appendix of (23).

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